

Ayushman Kumar

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GitHub | LinkedIn | Portfolio | LeetCode

Education

- **IIT Patna** – B.Tech Artificial Intelligence & Data Science Aug 2024 – Jul 2028
- **CPI: 9.10/10** (4 semesters) *Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networking*
- **M.D.J Public School** – Higher Secondary (PCM): **96%**, Class X: **97.5%**

Achievements

- **Competitions:** 2nd Place – JILO India Health Hackathon (2026) | 3rd Place – Droidrun DevSprint (2026)
- **Academics & Problem Solving:** LeetCode: 500+ problems solved | JEE Advanced AIR 6013 | CBSE Class XII District Topper

Projects

- **Distributed Background Job Processing Engine** May 2026 – June 2026
 - *Tech Stack:* Go, Redis, Lua, Prometheus, Grafana, Docker
 - Architected a scalable distributed platform applying **SOLID principles** and Go concurrency via a customized worker pool to process over **8,000+ jobs/minute**.
 - Engineered strict at-least-once delivery by deploying atomic queues via **BRPopLPush**, safeguarding multi-threaded tasks against sudden worker node crashes; built a Sweeper service for fault tolerance.
 - Profiled system performance utilizing **Go pprof** and wrote Lua scripts to eliminate race conditions, reducing Redis round-trips by **40%**; achieved **90%+ unit test coverage** ensuring reliable execution.
- **Clinical Data Normalization Engine** – JILO India Health Hackathon Feb 2026 – Mar 2026
 - *Tech Stack:* Python, LangChain, React, PostgreSQL, pgvector, FHIR, RAG, LLMs
 - Built end-to-end **Machine Learning pipelines** processing unstructured diagnosis reports, lab results, and discharge summaries into FHIR-compliant patient records.
 - Developed a grounded RAG system leveraging ICD-10 and CPT knowledge bases to automate medical coding and populate **50+ insurance claim fields**.
 - Achieved **87% clinical entity extraction accuracy** while minimizing hallucinations, reducing pre-authorization processing time from **2+ hours to under 8 minutes**.
- **SecureVote – Decentralized Voting Platform** Aug 2025 – Sept 2025
 - *Tech Stack:* MERN Stack, OAuth 2.0, Linkable Ring Signatures
 - Designed a secure, decentralized cryptographic protocol relying on Linkable Ring Signatures to guarantee absolute voter anonymity alongside mathematical double-vote prevention.
 - Developed custom Role-Based Access Control (RBAC) middleware modules to handle deterministic state management across **6 complex lifecycle stages** of an active election.
 - Optimized verification latency to execute sub-100ms algorithmic checks per cast ballot while maintaining verifiable audit trails. Demo | GitHub

Skills

- **Languages:** Python, JavaScript (ES6+), C/C++, SQL, Go, Bash/Shell
- **Backend & Data:** Node.js, Express.js, FastAPI, PostgreSQL, MongoDB, Redis, Prisma ORM, pgvector
- **Infrastructure & Core:** Linux/Unix, Docker, Kubernetes (K3s), Nginx, CI/CD, Object-Oriented Design (OOD)
- **AI & Systems:** LangChain, PyTorch, Machine Learning pipelines, RAG, HNSW indexing, semantic search, NLP
- **Security:** Cryptographic protocols, Linkable Ring Signatures, OAuth 2.0, RBAC, session synchronization

Leadership Experience

- **Head, Web & Media Wing – Academic Career Council (ACC), IIT Patna:** Collaborated across student bodies and administrative teams to manage web infrastructure for acc.iitp.ac.in; scaled platform architecture to support **3,000+ active students**.
- **Web Development Lead – Google Developer Groups (GDG), IIT Patna:** Actively supported peers and fostered an engineering culture of productivity by hosting **8+ deep-dive technical events** and mentoring **20+ members** in advanced system architecture design.
- **Flagship Coordinator – Celesta (IIT Patna Tech Fest):** Managed cross-functional operational logistics and synchronization between multiple sub-teams, handling technical scheduling for **100+ national participants**.